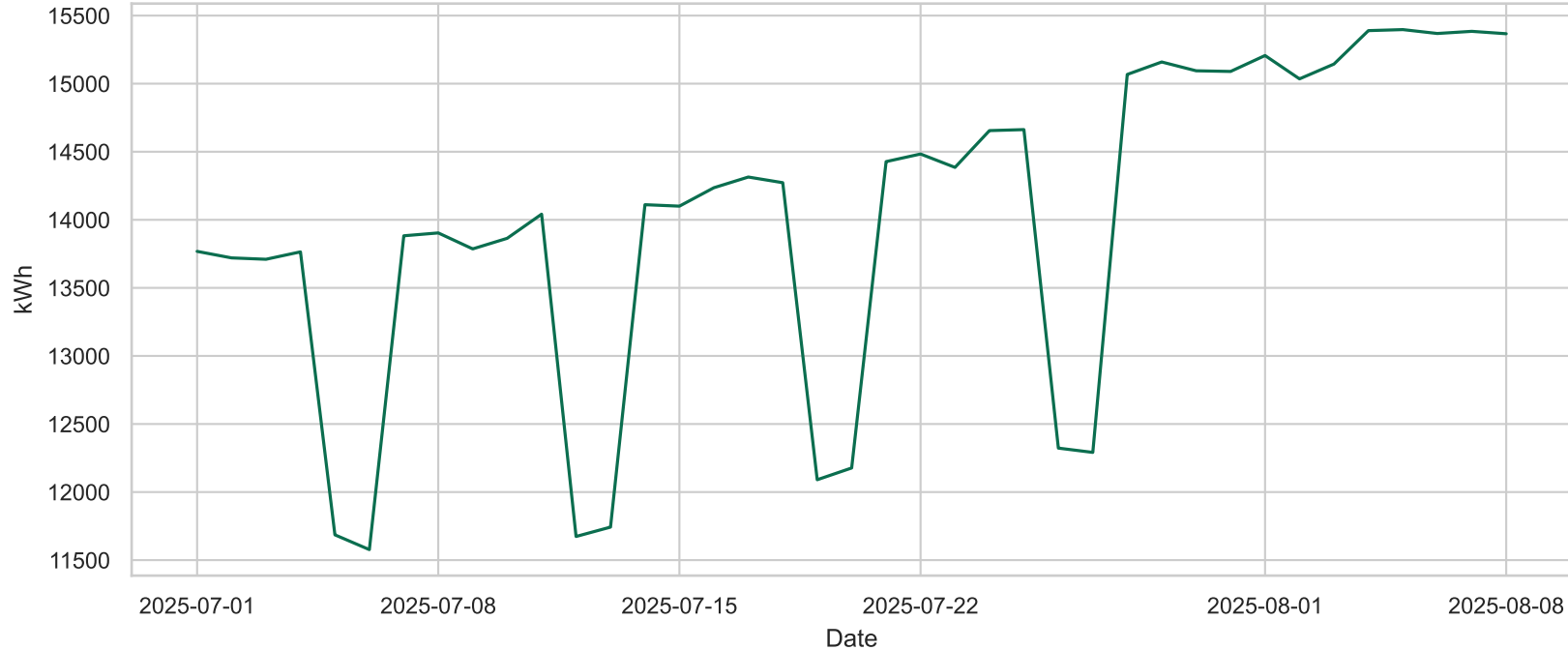
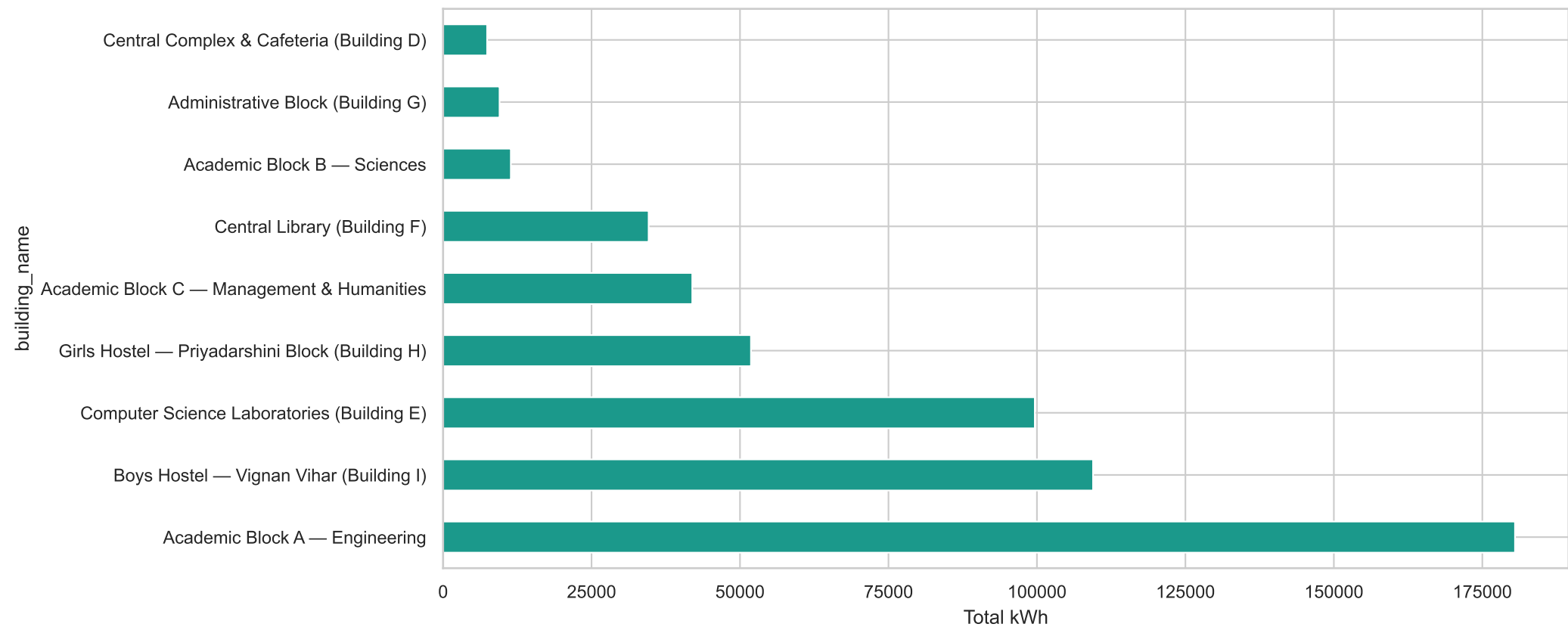


1. Daily campus energy trend



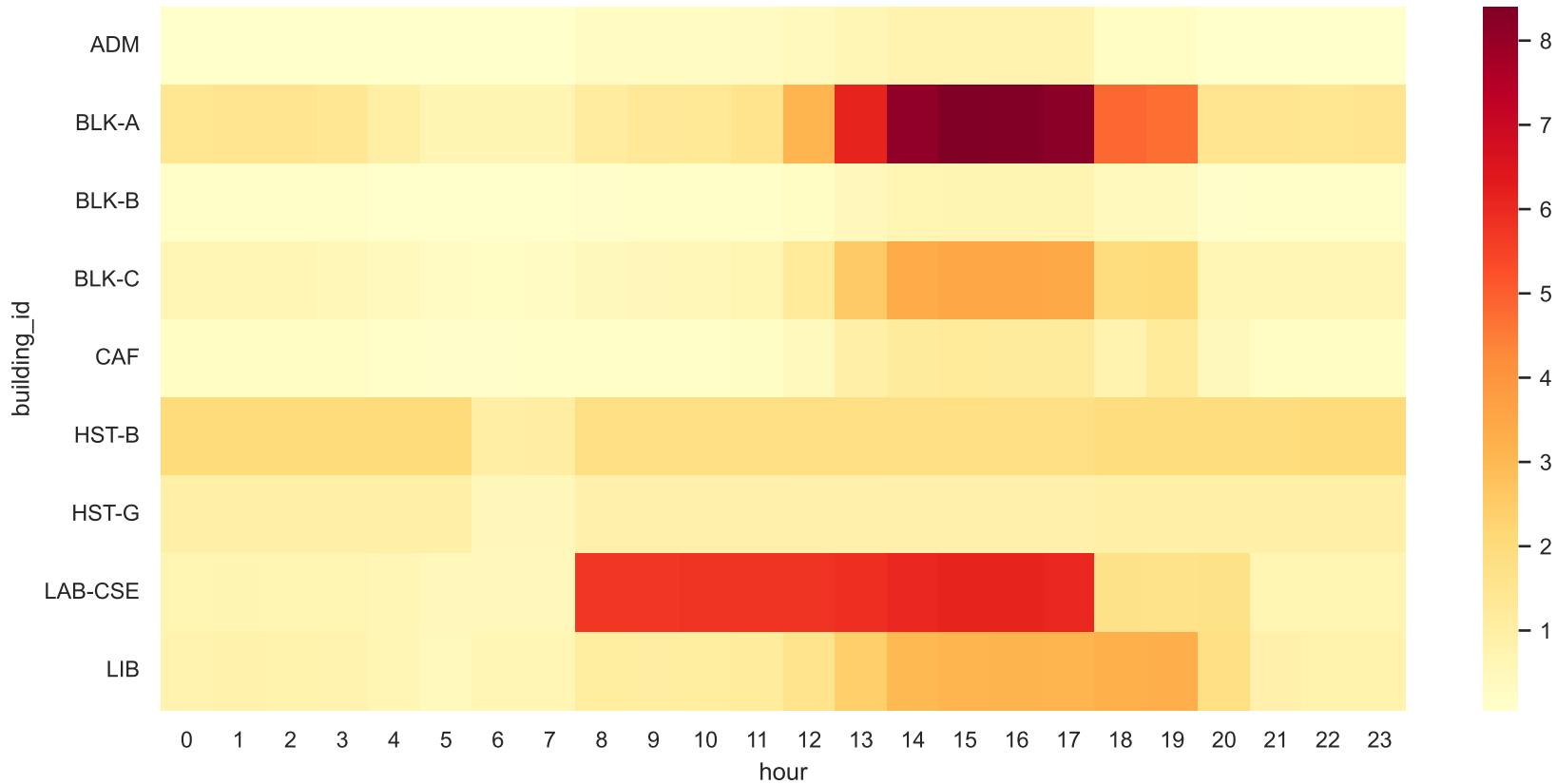
Observation: Campus load shows weekday peaks and exam-period lift in library/lab demand.

2. Building-wise consumption



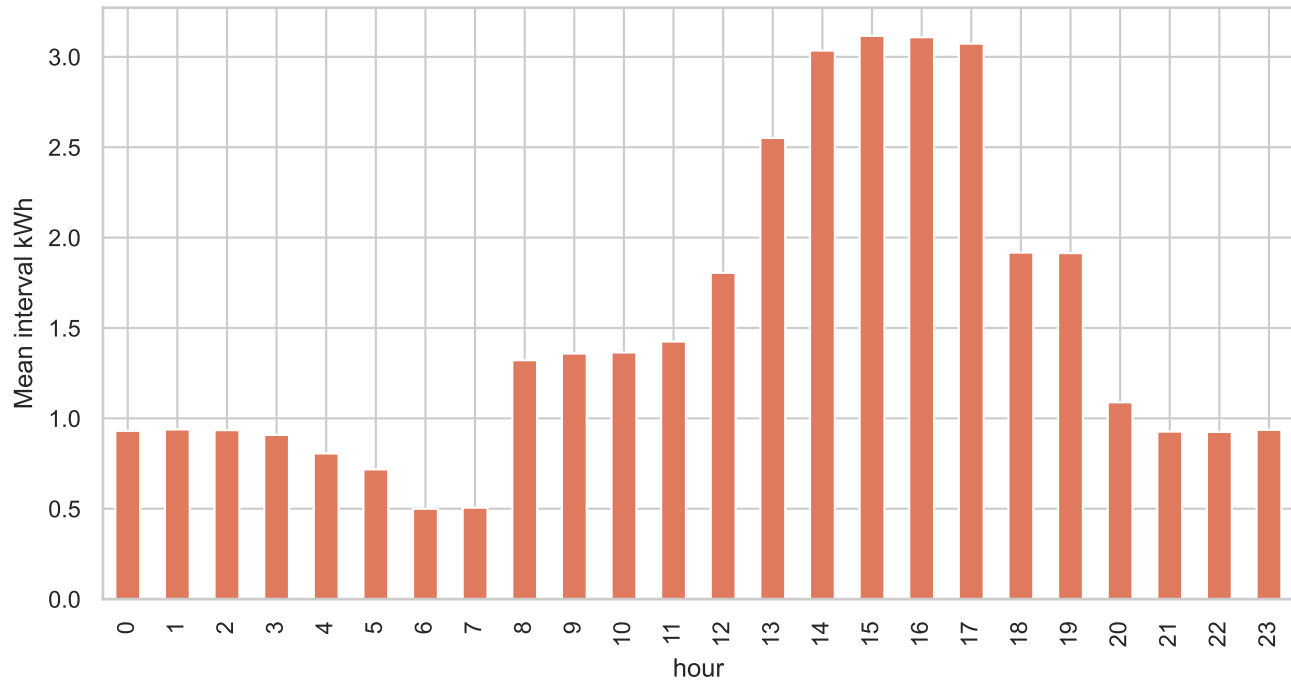
Observation: Hostels and computer labs dominate total energy due to occupancy duration and IT/HVAC load.

3. Building × hour energy heatmap



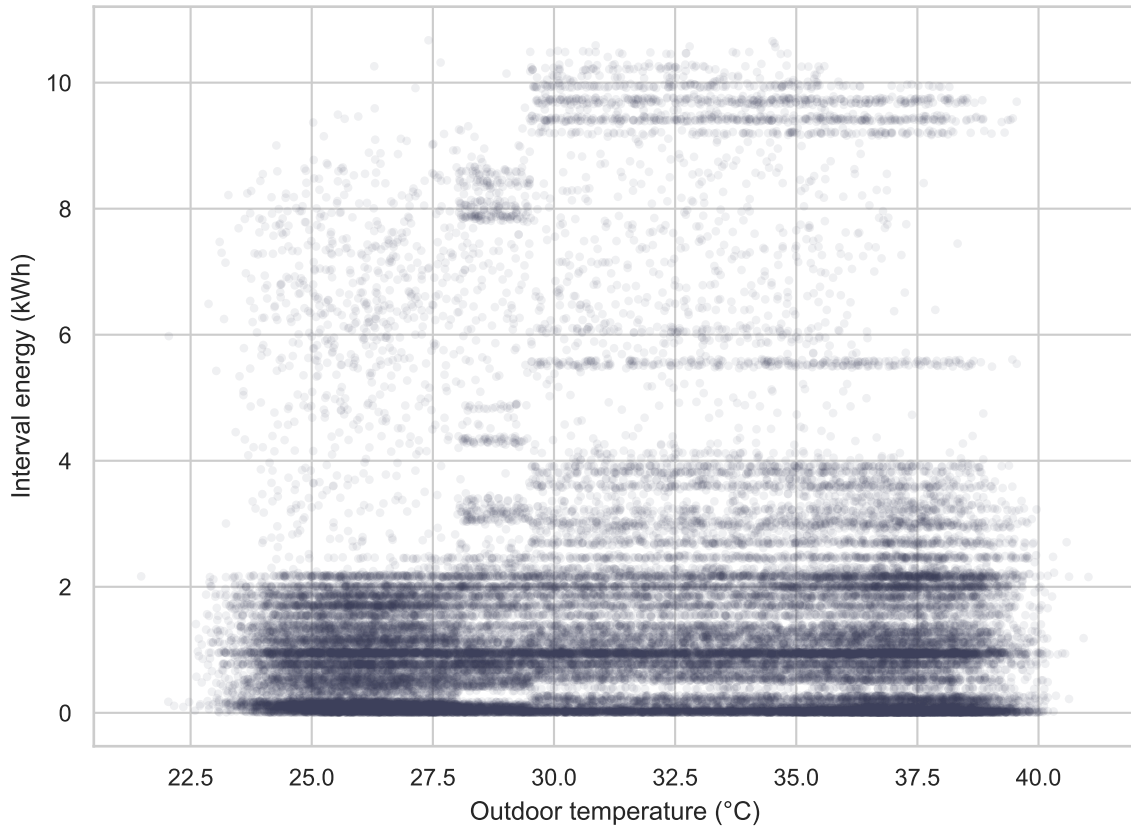
Observation: Academic blocks peak 09:00–16:00; hostels invert with night-time occupancy.

4. Peak hour analysis



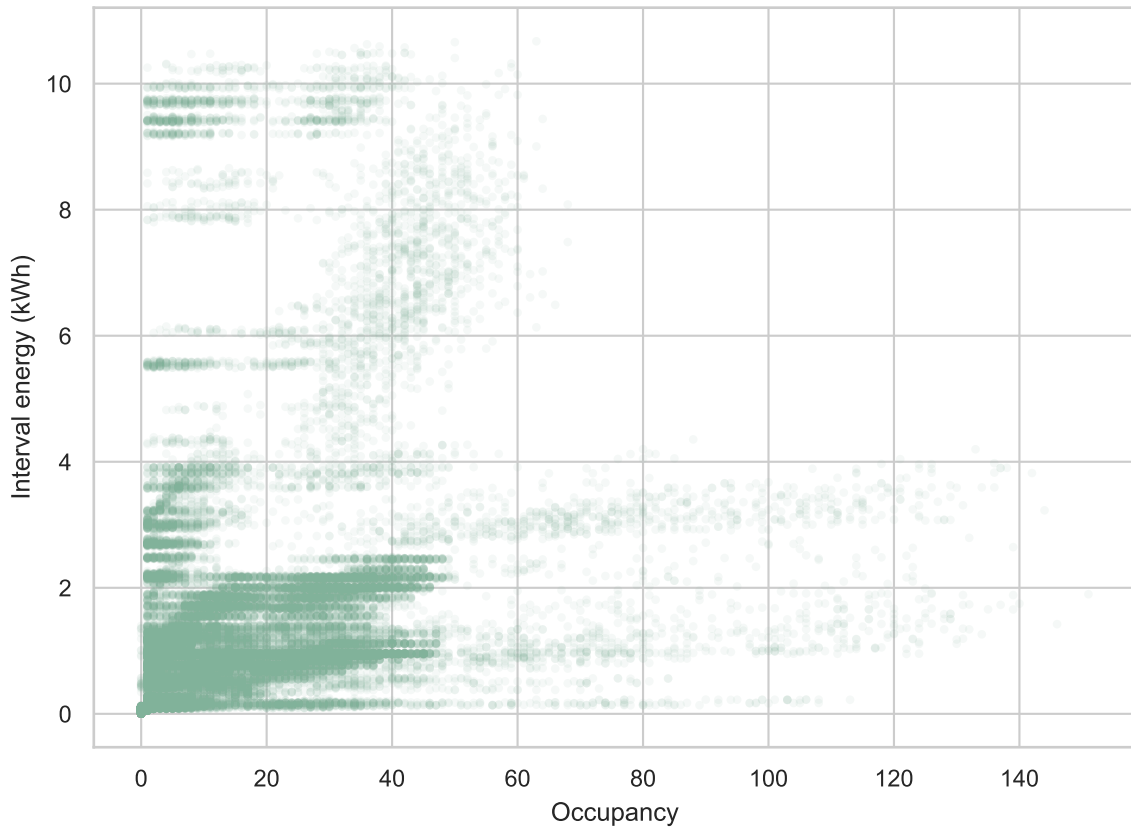
Observation: Mean interval energy rises through late morning and remains elevated until evening hostel load.

5. Weather vs energy



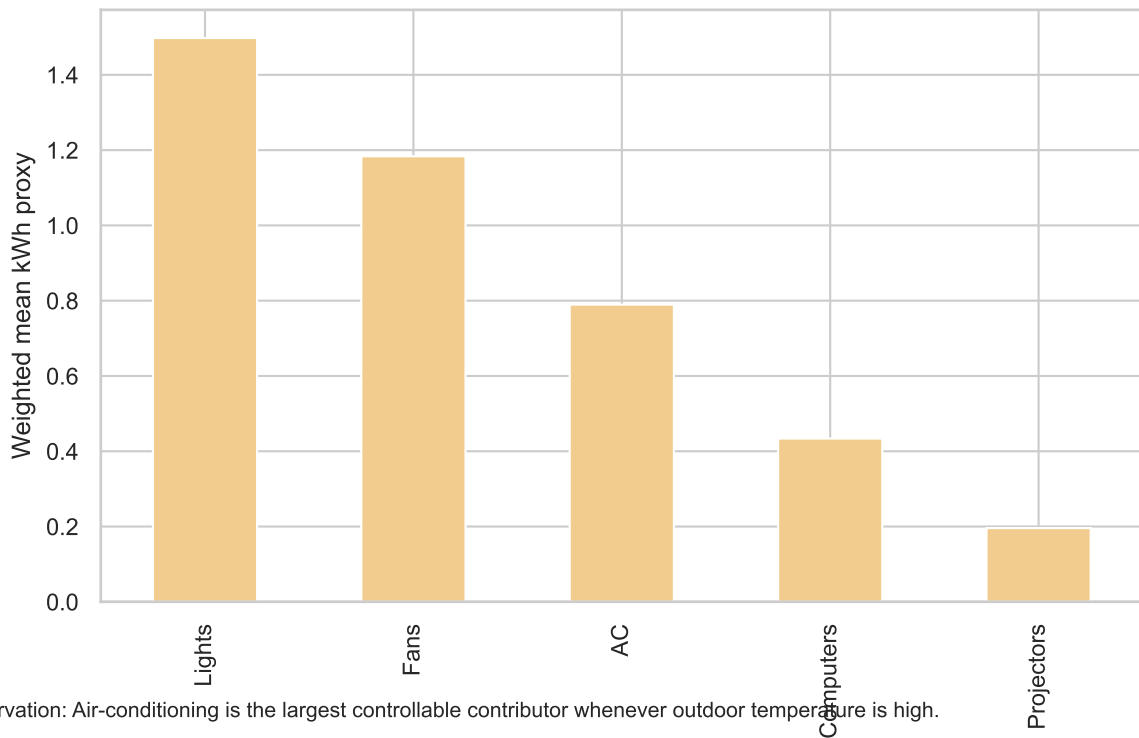
Observation: Higher outdoor temperature correlates with AC-driven energy, especially above $\sim 29^{\circ}\text{C}$.

6. Occupancy vs energy

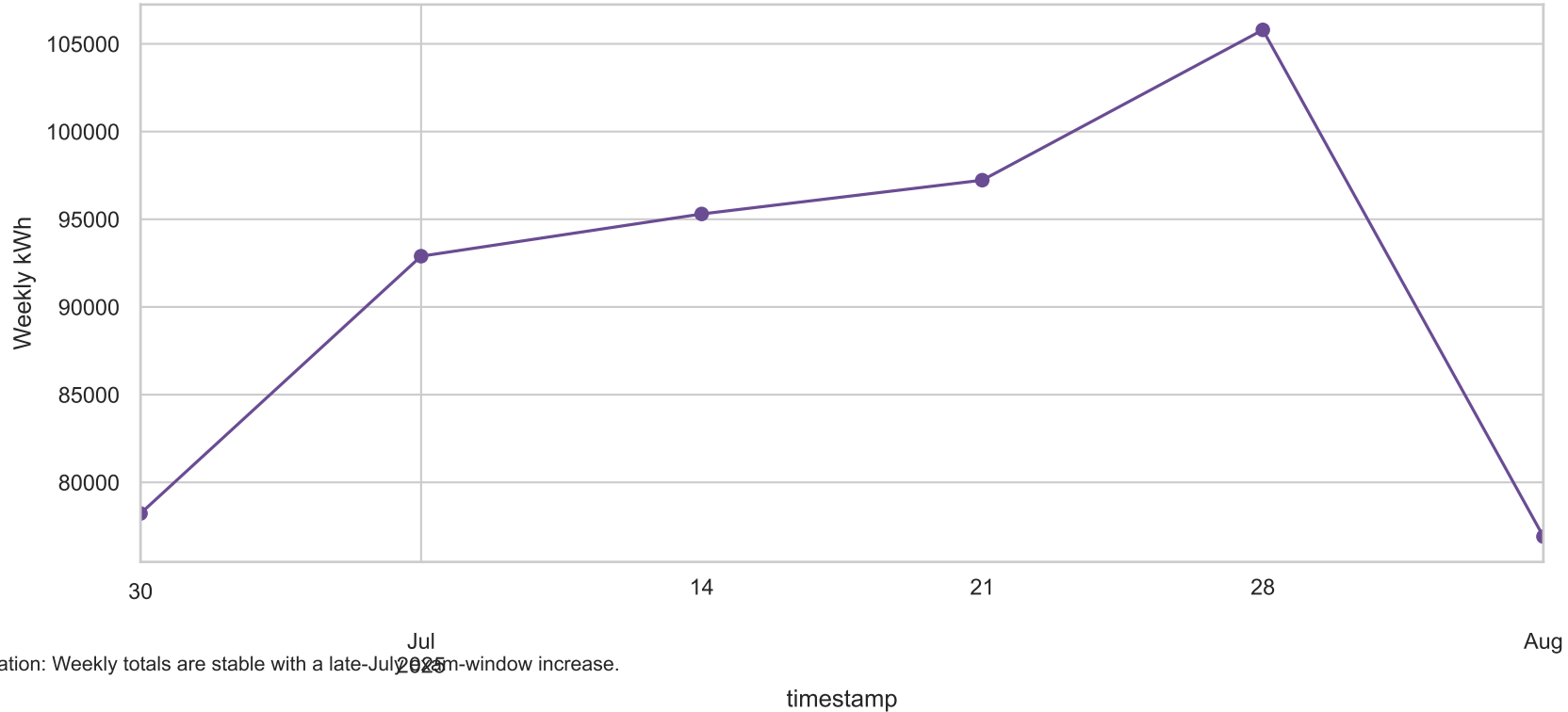


Observation: Energy scales with occupancy but HVAC/IT rooms sit above the occupancy-only trend.

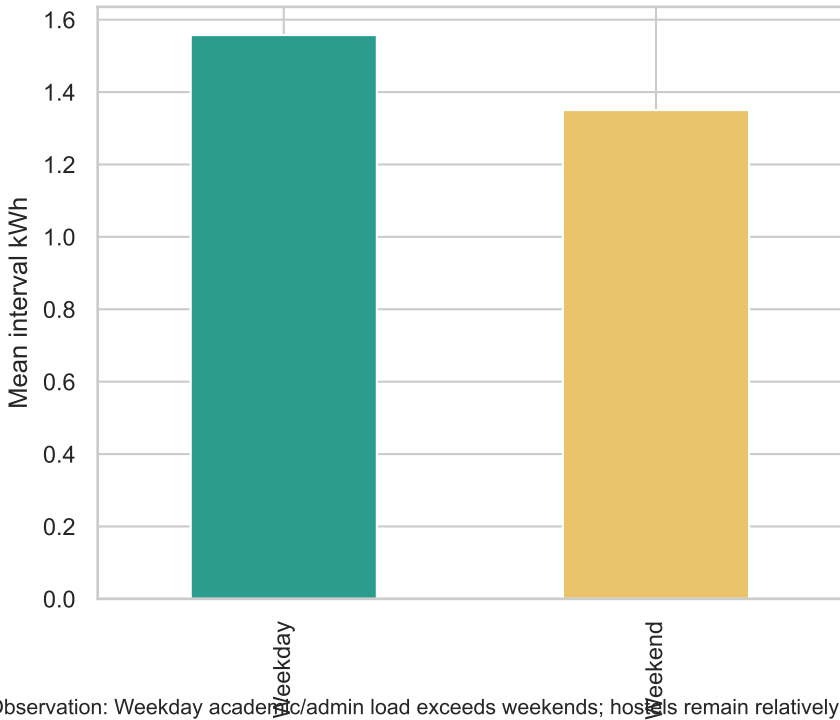
7. Device contribution proxy



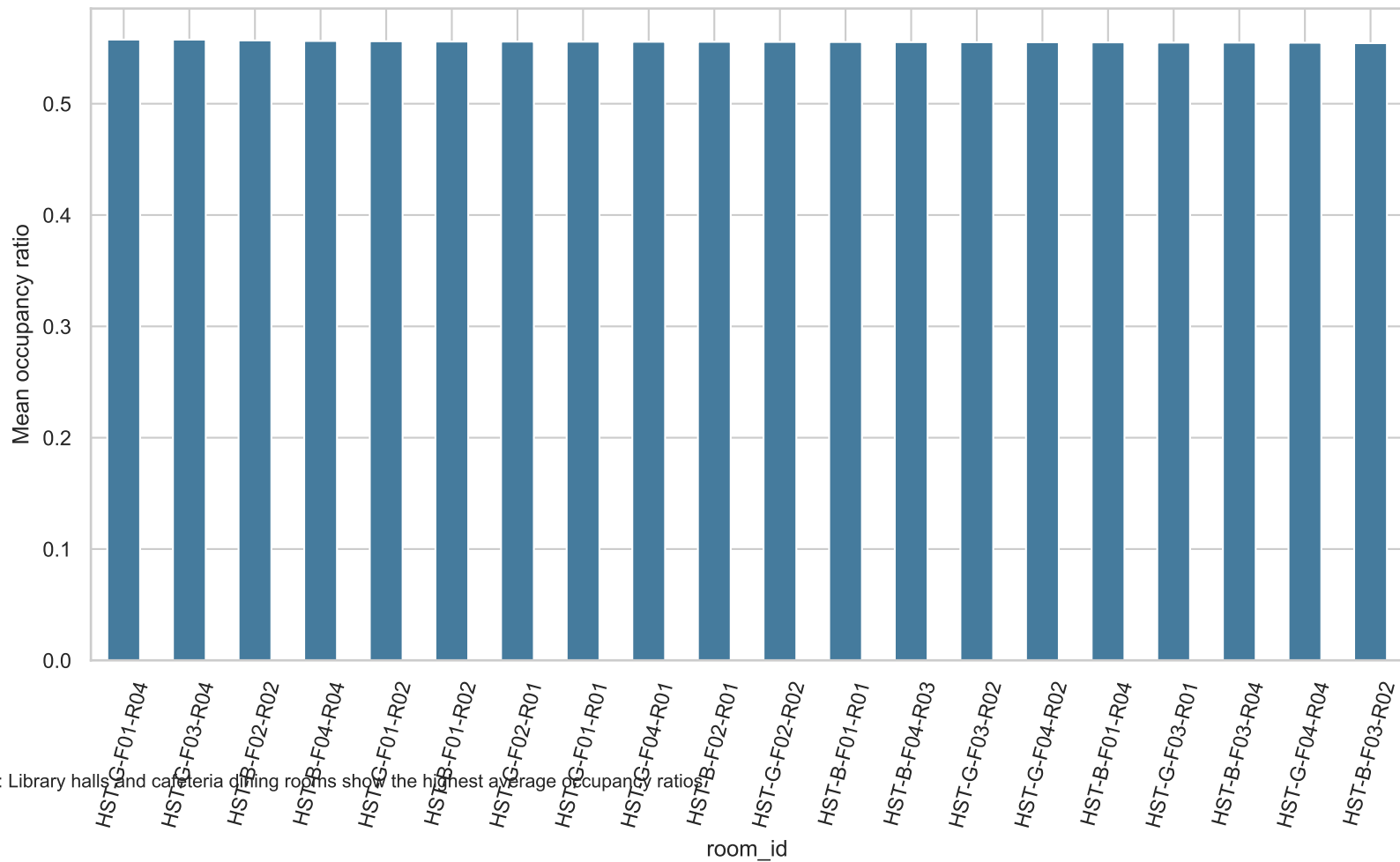
8. Weekly / monthly-scale trend



9. Weekend vs weekday

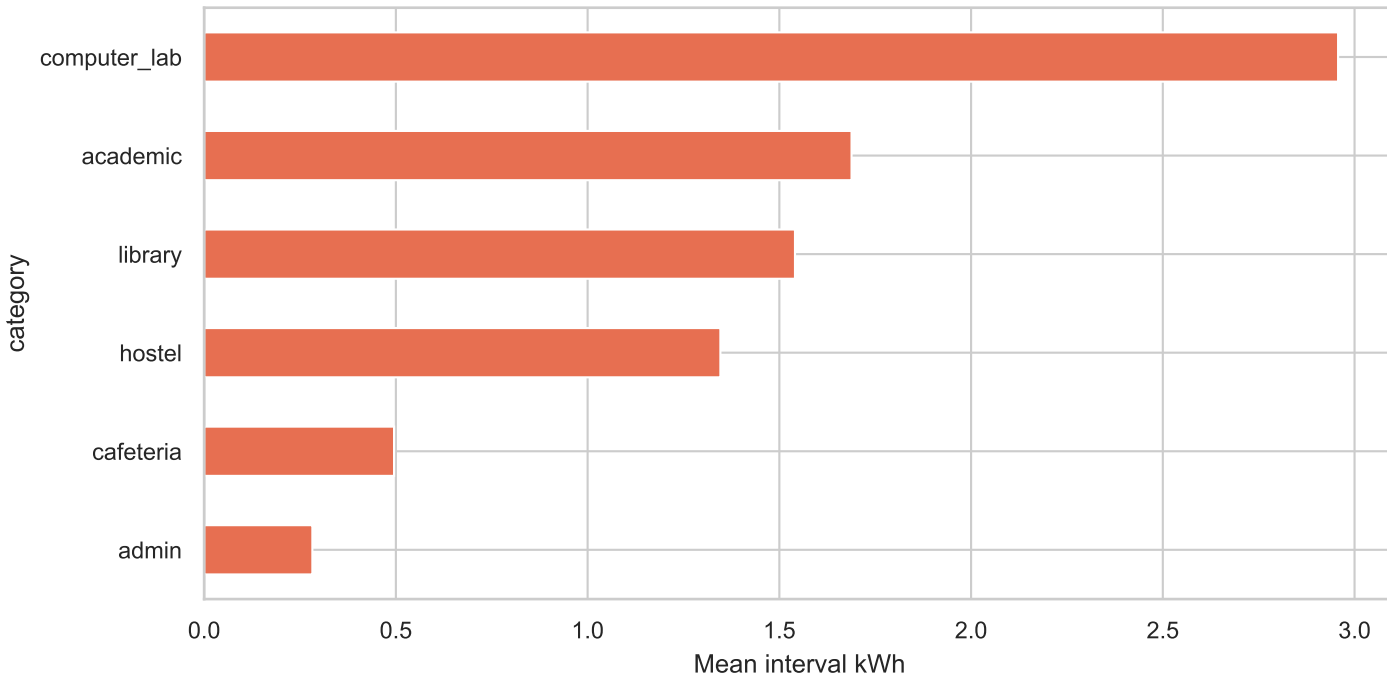


10. Top-20 room utilization



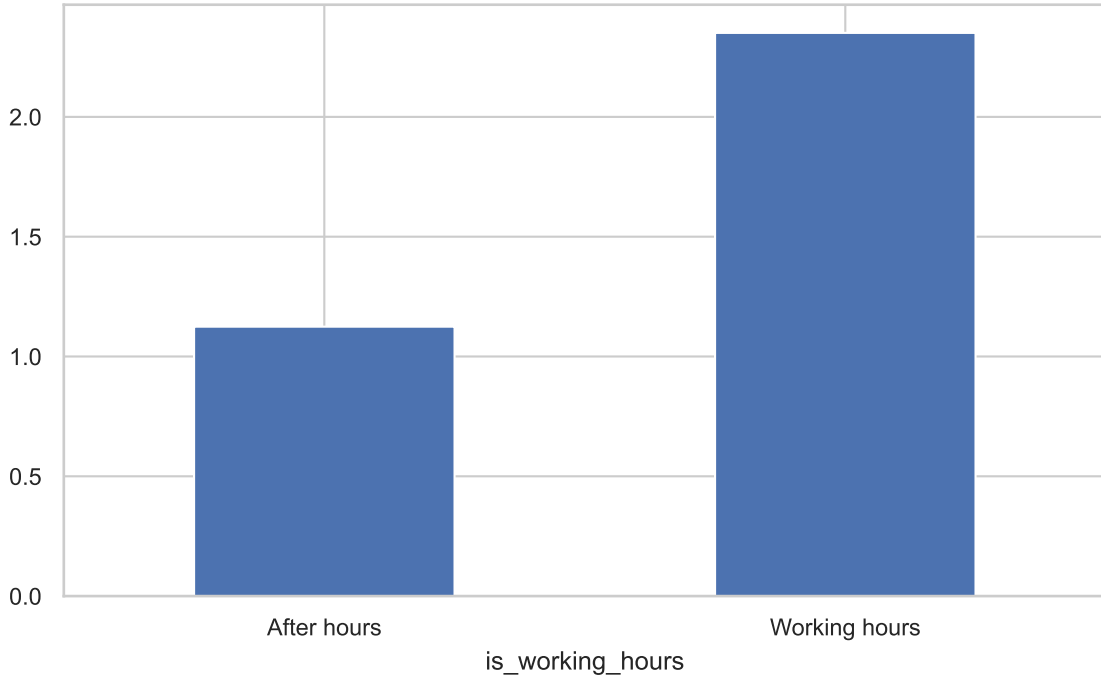
Observation: Library halls and cafeteria dining rooms show the highest average occupancy ratios.

11. Category mean energy intensity



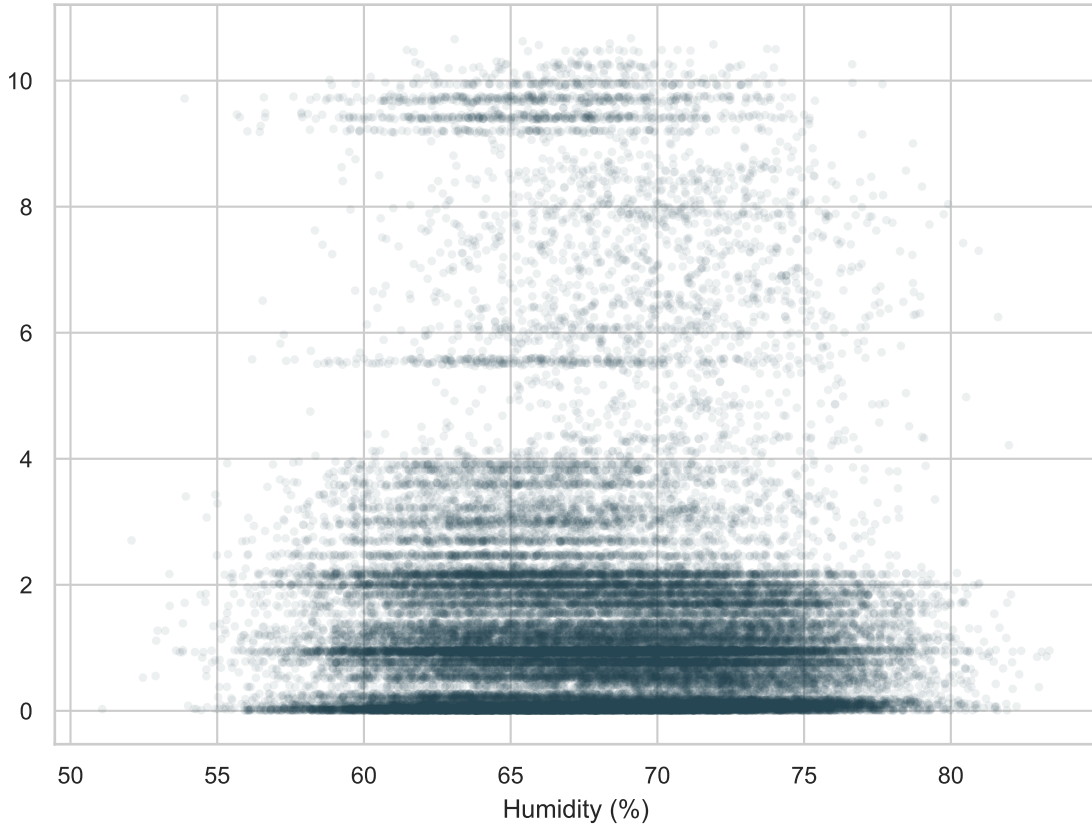
Observation: Computer labs have the highest mean intensity due to dense IT load plus HVAC.

12. Working-hours energy split



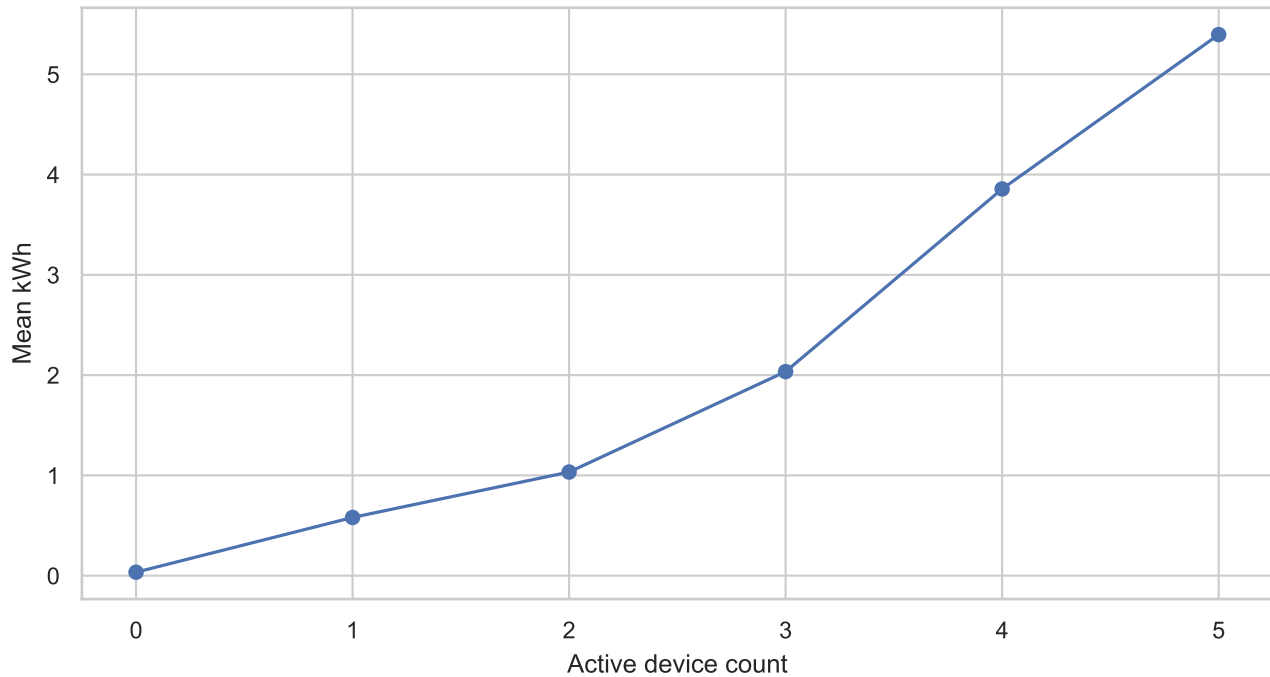
Observation: Working hours raise academic energy; after-hours residual load is a wastage target.

13. Humidity vs energy



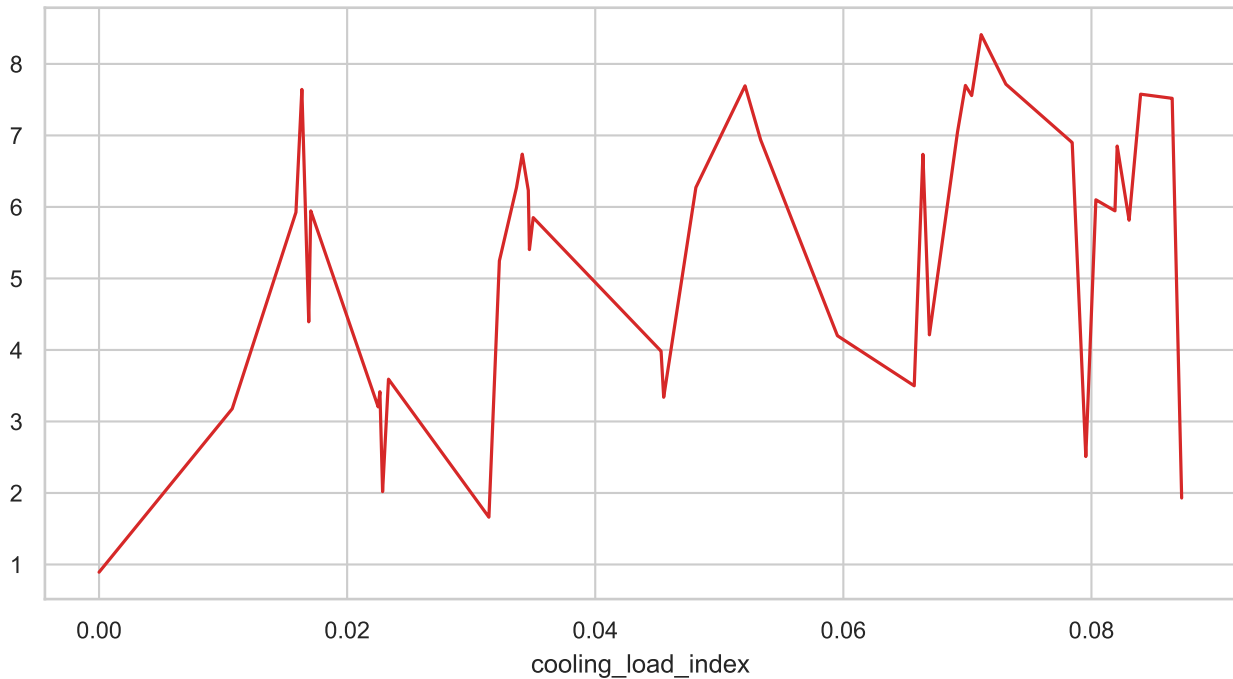
Observation: Humidity is weakly associated with energy; temperature and occupancy dominate.

14. Active devices vs energy



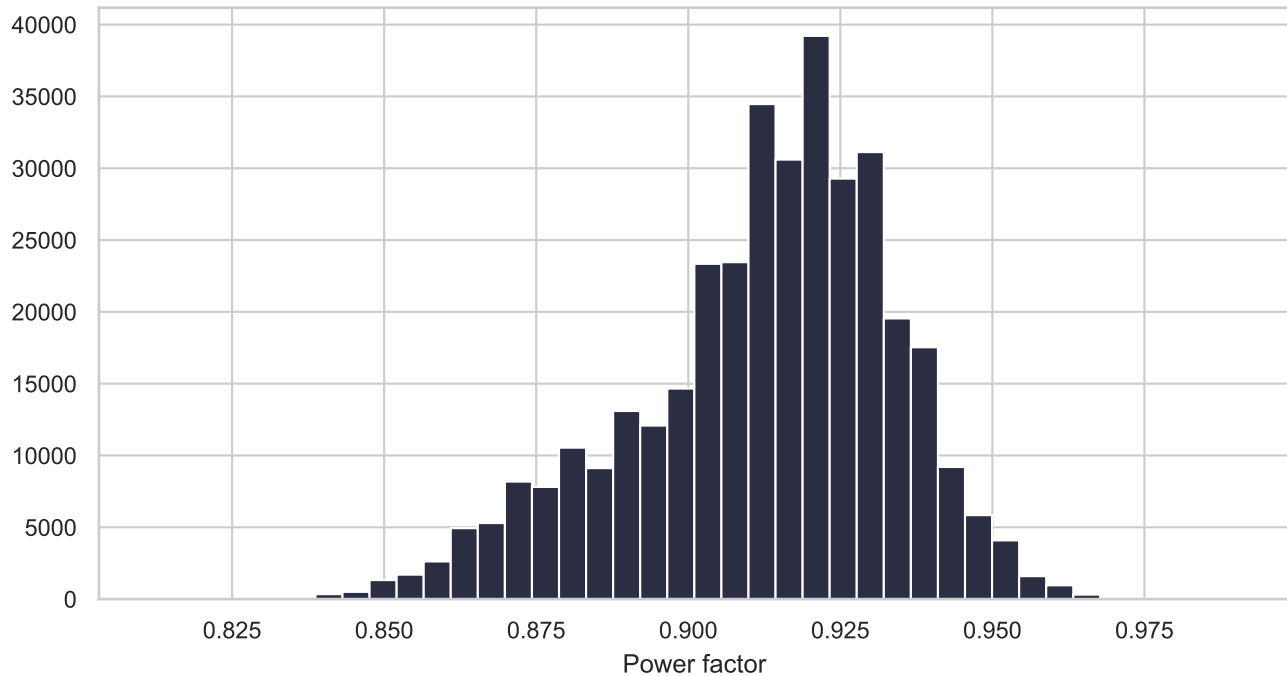
Observation: Mean energy increases monotonically with concurrently active end-use devices.

15. Cooling load index vs energy



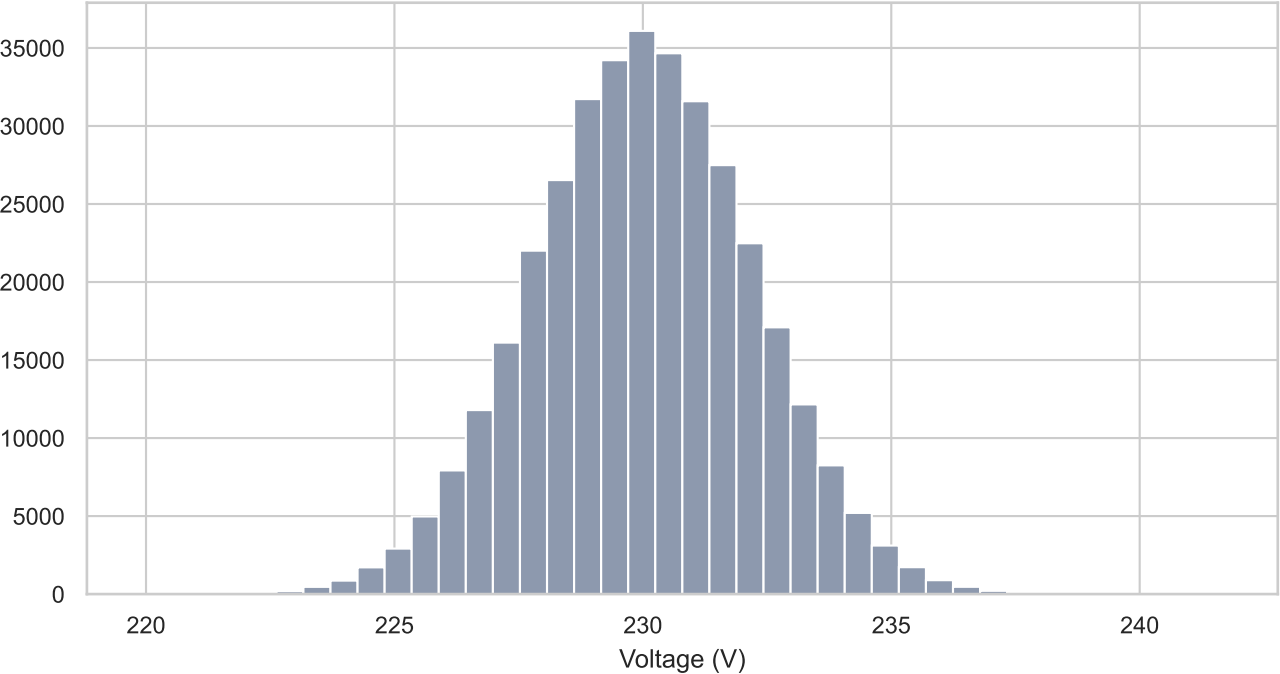
Observation: Cooling load index is a strong engineered predictor of HVAC-dominated intervals.

16. Power factor distribution



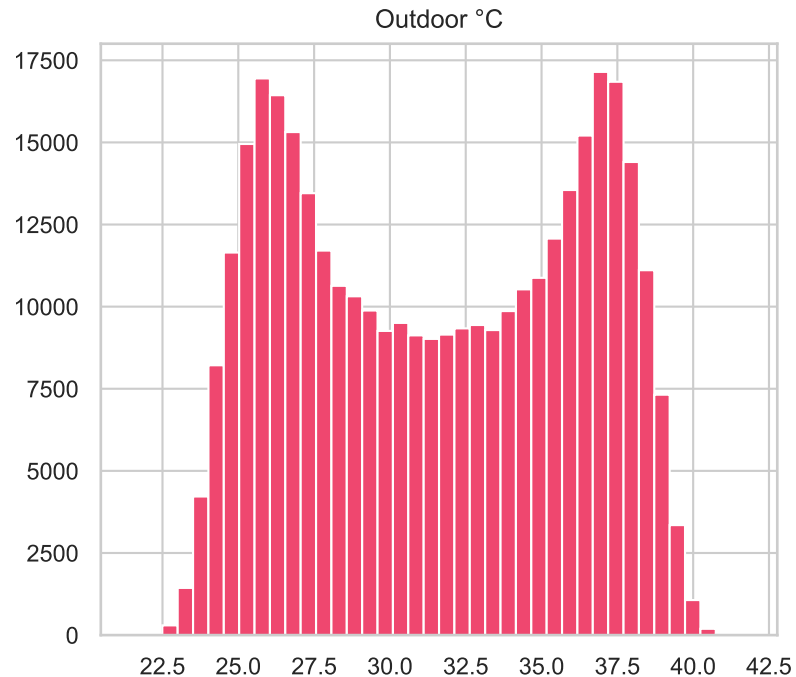
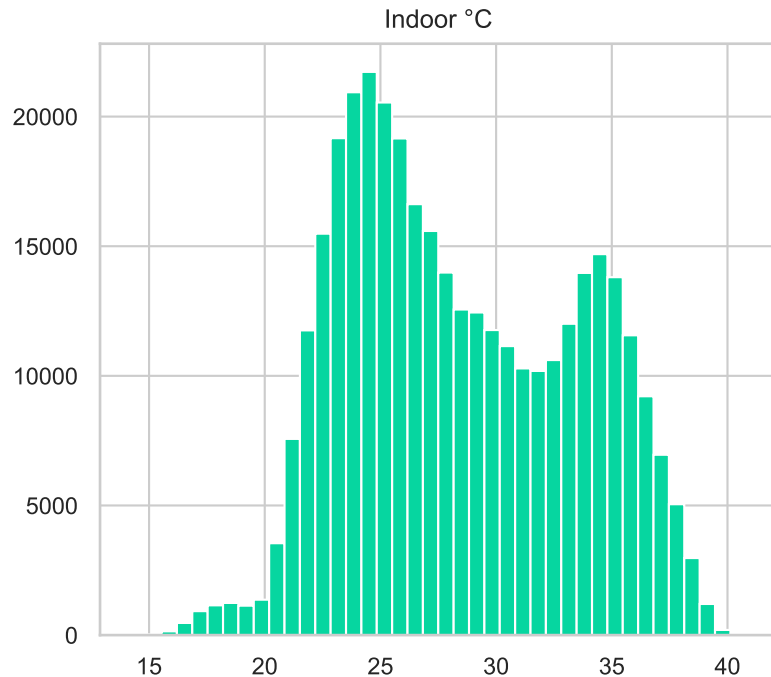
Observation: Power factor clusters near 0.90–0.96; AC-on intervals pull PF slightly lower.

17. Voltage distribution



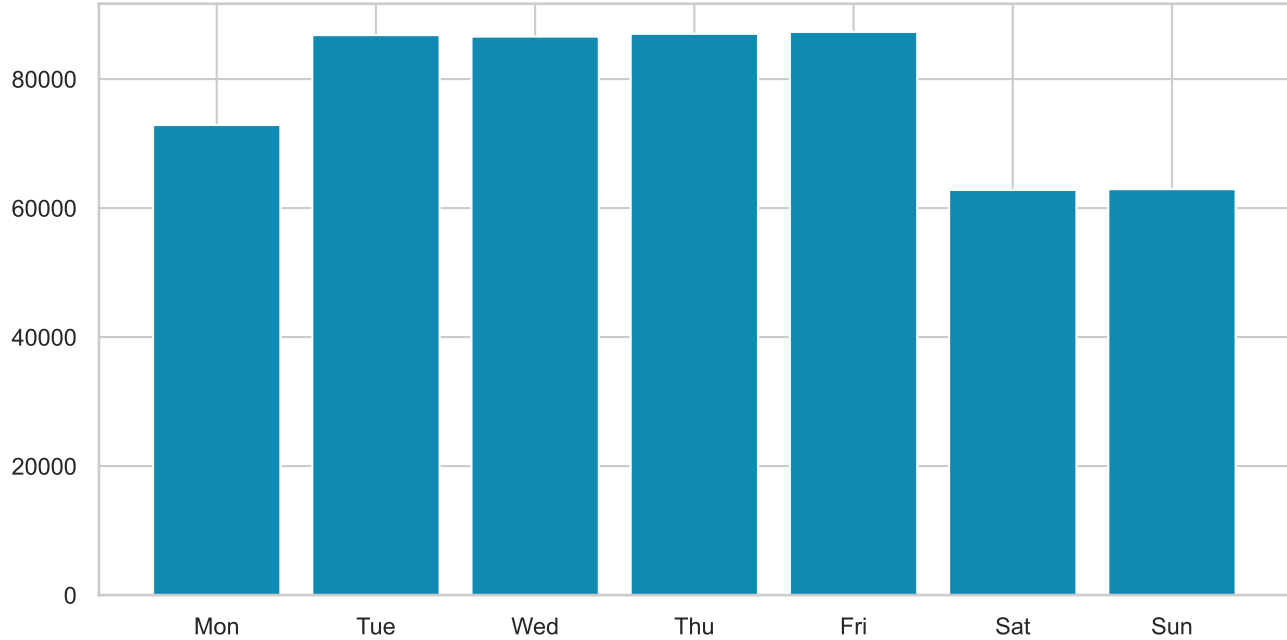
Observation: Branch voltage stays around the 230 V Indian nominal with realistic feeder noise.

18. Indoor vs outdoor temperature



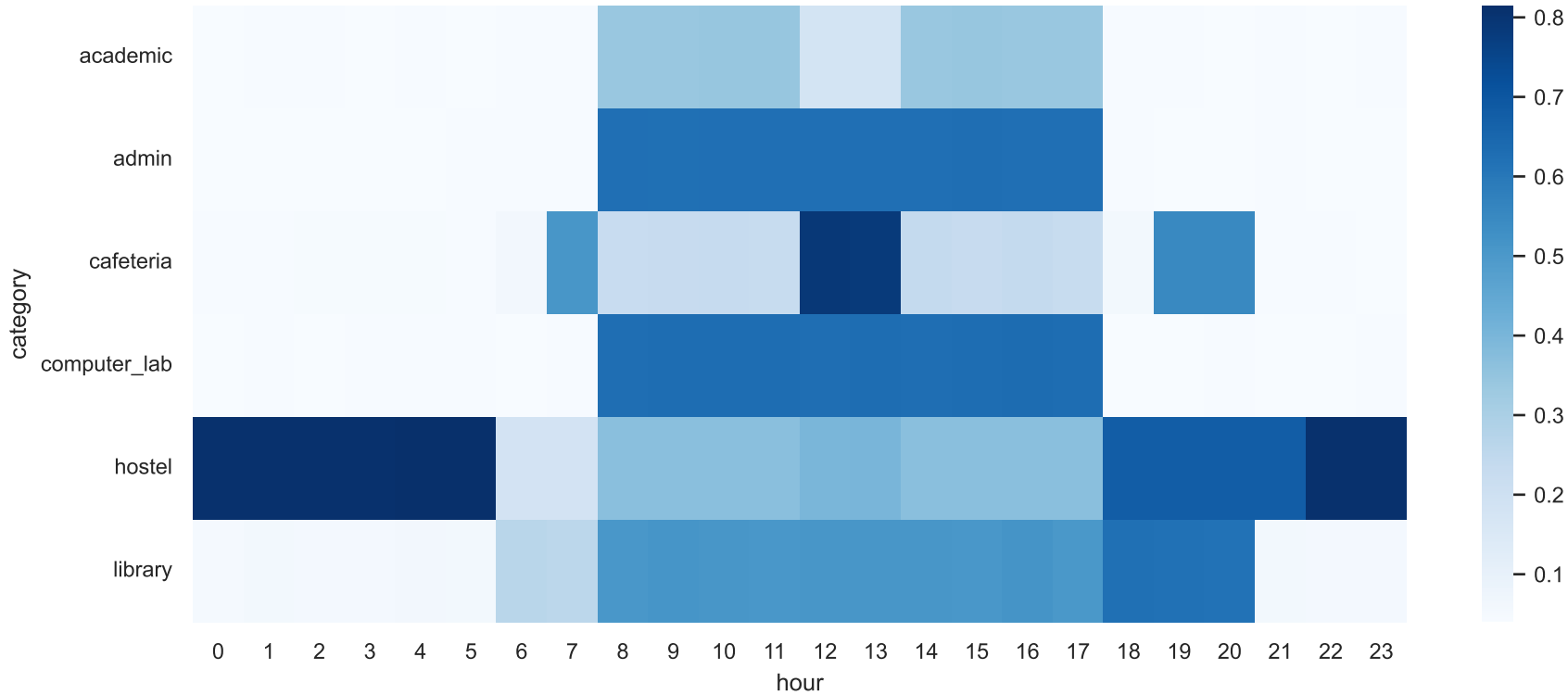
Observation: Indoor temperatures are suppressed relative to outdoor when HVAC is active.

19. Day-of-week energy



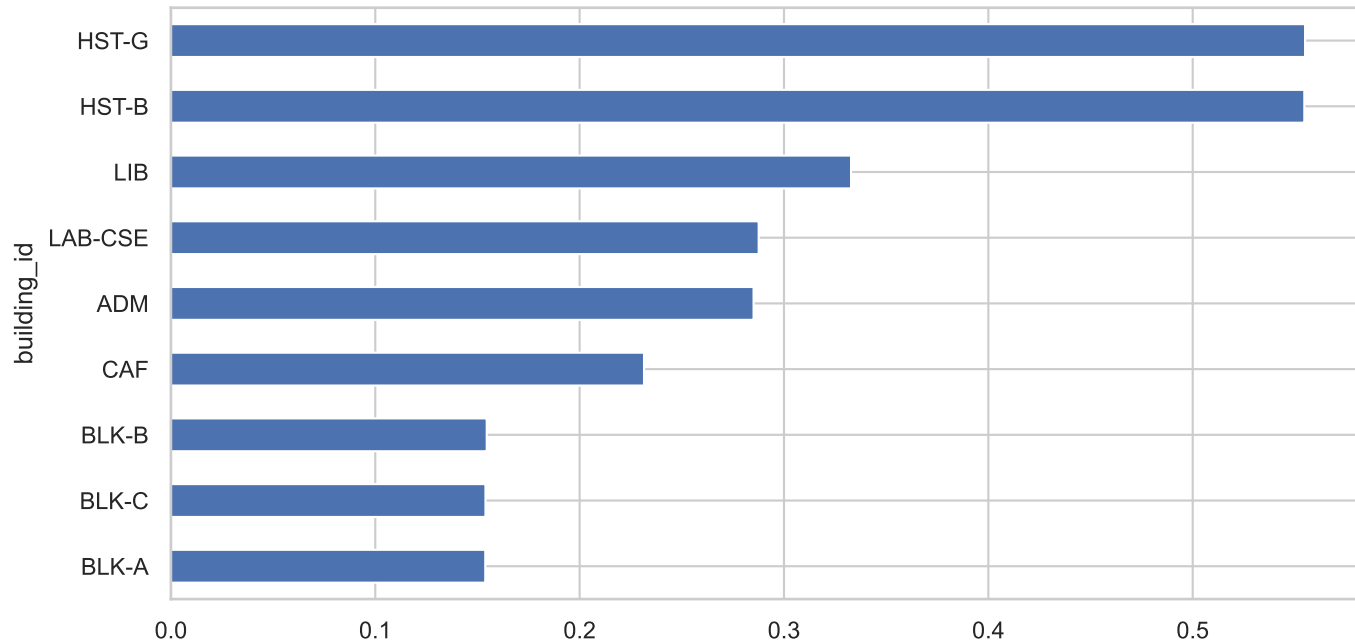
Observation: Monday–Friday academic schedules produce higher campus totals than weekends.

20. Occupancy heatmap by category



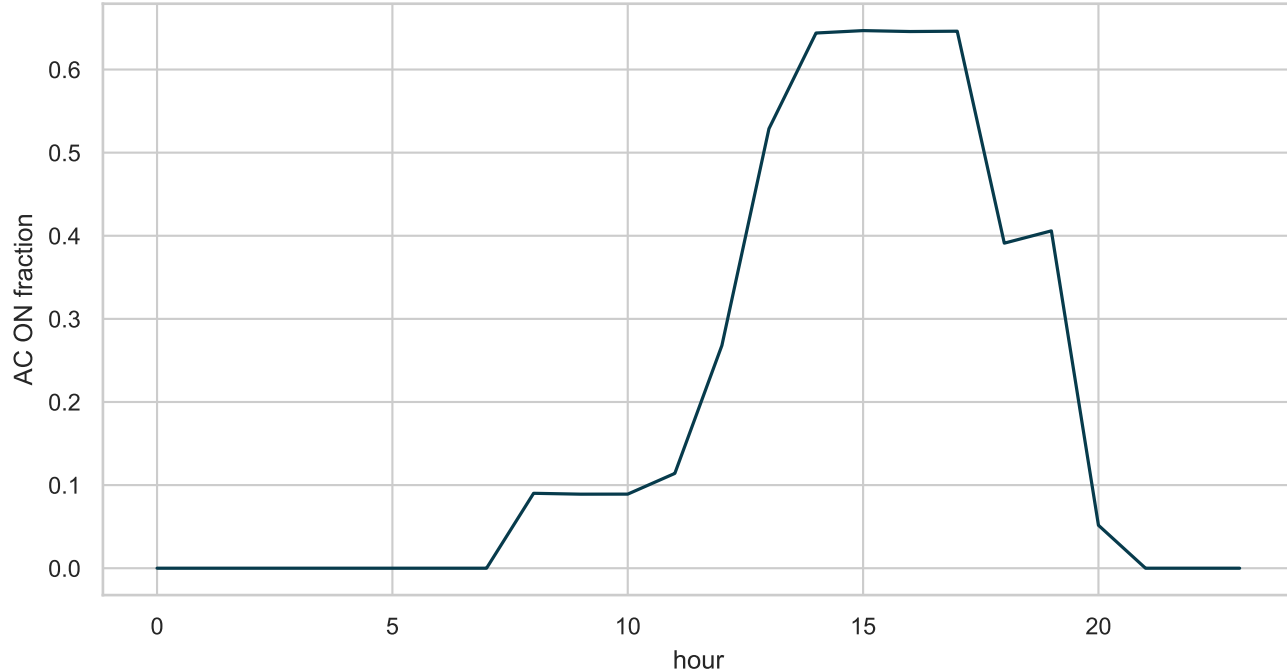
Observation: Hostels fill overnight; cafeteria peaks at lunch; library occupancy rises in the exam window.

21. Mean building utilization



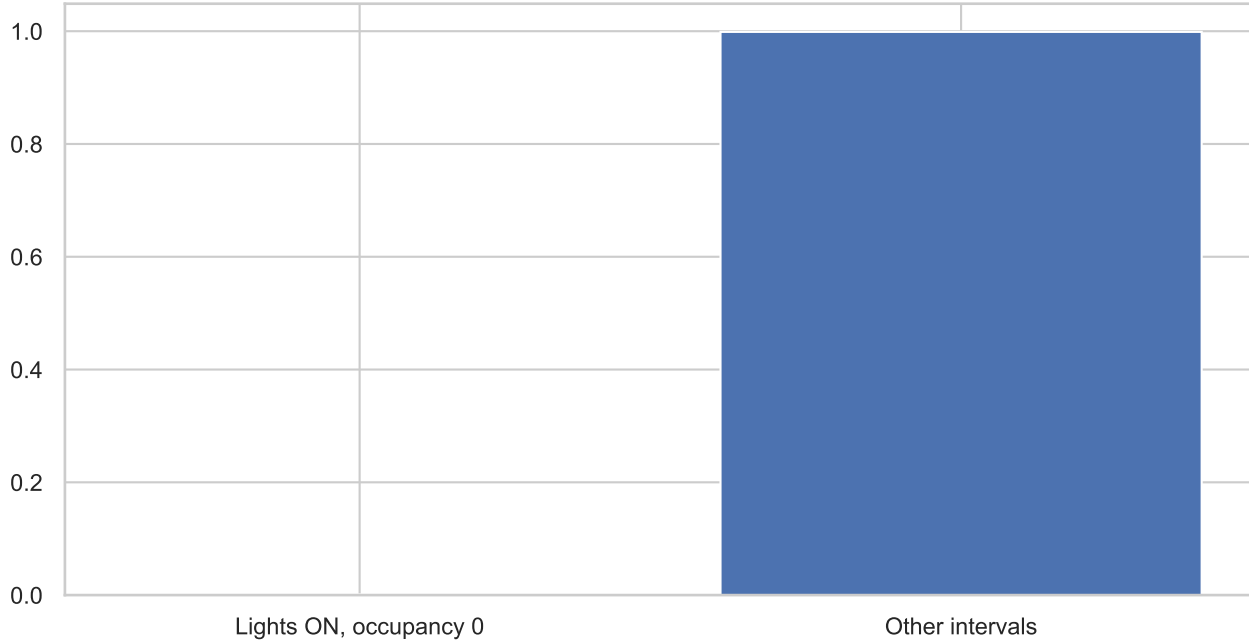
Observation: Admin utilization is high during workdays but low absolute energy due to smaller rooms.

22. AC operating profile



Observation: AC duty cycle tracks hot working hours — the primary HVAC optimization lever.

23. Empty-room lighting wastage share



Observation: About 0.12% of intervals show lights on with zero occupancy — a lighting-control target.

24. Building weekend vs weekday intensity

